

PROBLEM 6

Churn Prediction

Identifying customers who are likely to leave before they actually leave

ML Interview Preparation Guide

What Is Churn Prediction?

Churn is when a customer stops using your product or service. Acquiring a new customer costs 5-7x more than retaining an existing one. A 5% reduction in churn can increase profits by 25-95%. By the time a customer cancels, it is usually too late to save them. Churn prediction means identifying who is about to leave so you can intervene early.

Real World Examples: Telecom (predicting subscription cancellations), Spotify/Netflix (predicting non-renewals), banks (predicting account closures), games (predicting players about to quit), SaaS (predicting business customers not renewing contracts).

Step 0: Clarify the Problem

- **What is churn?** Canceling a subscription (hard churn)? Not logging in for 30 days (soft churn)? Downgrading from paid to free (revenue churn)? Each definition gives different training labels!
- **Prediction horizon?** Predict churn in next 7 days? 30 days? 90 days? Shorter = more accurate but less time to intervene.
- **What action do we take?** Discount? Personal outreach? We only want to intervene on customers who would respond — offering a discount to someone staying anyway wastes money.
- **B2B or B2C?** B2B: fewer customers, higher value, richer behavioral data. B2C: millions of customers, lower value, noisier behavior.

Step 1: Understanding the Data

Defining Your Observation and Label Windows

This is the most technically important part. Observation window: Jan 1 to Jan 31 — collect all features. Label window: Feb 1 to Feb 28 — did they cancel? You must NEVER use information from the label window when computing features. This is data leakage — the #1 mistake in churn modeling.

Types of Churn

- **Contractual churn (SaaS, telecom):** Customer explicitly cancels. Clear label. Easy to define.
- **Non-contractual churn (e-commerce, apps):** Customer just stops coming. No explicit cancellation. You define it yourself — e.g., no purchase in 90 days.

Survival Analysis — A Different Way to Think

Instead of "will this customer churn in 30 days?" ask "how long will this customer stay?" The survival curve shows what fraction of customers are still active after t days. Key insight: early

churn is disproportionately high. Customers who do not see value in the first 30-60 days almost always leave. This is why onboarding is so critical.

Segment Your Customers First

Before modeling, look at churn rate by segment: Starter plan = 12% churn rate (very high!), Pro plan = 4%, Enterprise = 1% (very sticky). By tenure: 0-30 days = 20% churn, 31-180 days = 6%, 180+ days = 2%. By usage: 0 logins/week = 35% churn, 5+ logins/week = 1%. This EDA alone tells you where to focus interventions.

Step 2: Feature Engineering

Account Features

- `plan_type`, `contract_length`, `tenure_days`, `num_seats`, `mrr`
- `is_auto_renew` — turned off auto-renew? Strong churn signal!
- `days_until_renewal` — close to renewal = high-risk window

Product Usage Features (Most Predictive)

- `days_since_last_login` — Recency
- `logins_last_7d`, `logins_last_30d`, `active_days_last_30d` — Frequency
- `num_features_used_30d` — breadth of feature adoption (Depth)
- `num_integrations_enabled` — connected to other tools = stickier
- `num_collaborators_active` — team use = stickier product

Trend Features — The Most Powerful Signal

Raw numbers are not enough. 20 logins this month looks fine. But if they had 50 logins last month, that is a 60% drop — they are disengaging.

- `login_trend` = `logins_this_month` / `logins_last_month`
- `feature_usage_trend` = `features_used_this_month` / `features_used_last_month`
- `usage_slope_90d` — slope of usage over last 90 days via linear regression

The usage slope feature is extremely powerful. A customer who went from 30 logins/week to 5 over 3 months is very different from one who has been consistently at 5. Compute: `slope, intercept = polyfit(weeks, weekly_logins, 1)`. Negative slope = declining engagement = high churn risk.

Support and Health Features

- `num_support_tickets_30d`
- `num_unresolved_tickets` — open tickets = frustrated customer
- `avg_ticket_resolution_time` — slow support = unhappy customer
- `has_billing_issues` — failed payment = likely to churn

- `nps_score` — Net Promoter Score (0-10)

Step 3: Statistics and Math

Survival Analysis — Kaplan-Meier Estimator

$S(t)$ = probability of surviving (staying) past time t . $S(t)$ = product over all event times $t_i \leq t$ of $[1 - (d_i / n_i)]$ where d_i = number of churns at time t_i and n_i = number still active at time t_i . This gives you the survival curve. Censoring: some customers are still active when you run your analysis. Kaplan-Meier handles censored observations elegantly.

Cox Proportional Hazards Model

$h(t) = h_0(t) \times \exp(b_1 \times \text{login_frequency} + b_2 \times \text{tenure} + b_3 \times \text{plan_type} + \dots)$. The hazard rate $h(t)$ = probability of churning right now given survived until t . Coefficients are interpretable:

- $\text{beta_login_frequency} = -0.8$ means $\exp(-0.8) = 0.45$, so each login per week reduces churn risk by 55%
- $\text{beta_billing_issue} = 2.1$ means $\exp(2.1) = 8.2$, so a billing issue multiplies churn risk by 8.2x!

Expected Value of Intervention

Not all at-risk customers are worth saving. $EV = P(\text{churn without intervention}) \times P(\text{saves if we intervene}) \times CLV - \text{Cost of intervention}$. Example: 500 dollar/month customer, 3 years if retained. $CLV = 18000$. $P(\text{churn}) = 0.7$, $P(\text{saves}) = 0.3$, $\text{Cost} = 500$ (discount). $EV = 0.7 \times 0.3 \times 18000 - 500 = 3280$ net positive — intervene.

Uplift Modeling

Not every customer who would churn will be saved by your intervention. Four segments: Persuadables (would churn but intervention saves them — TARGET THESE), Sure Things (would not churn anyway — waste money), Lost Causes (will churn regardless — save resources), Do Not Disturb (happy customers who do not need contact).

$\text{Uplift} = P(\text{retained} \mid \text{intervention}) - P(\text{retained} \mid \text{no intervention})$. Train two models: one on customers who received intervention, one on those who did not. $\text{Uplift score} = \text{Model_T.predict} - \text{Model_C.predict}$.

Step 4: Models

Logistic Regression

Interpretable baseline. In regulated industries (banking, insurance), you need to explain every decision. Logistic regression gives clear coefficients. "We flagged this customer because their login frequency dropped 60% and they have 3 open support tickets."

Random Forest

Ensemble of decision trees. Each tree sees a random subset of features and data. Final prediction = average across all trees. Averaging 100 diverse trees cancels out individual errors (variance reduction). Robust, handles non-linear relationships, built-in feature importance.

XGBoost / LightGBM — The Standard

Sequential boosting where each tree corrects mistakes of previous ones. Virtually always the best model for tabular churn data. Use `class_weight="balanced"` or `scale_pos_weight` to handle class imbalance. Early stopping prevents overfitting.

Survival Analysis Models

- **Kaplan-Meier:** Non-parametric, no features. Use for understanding overall customer lifecycle and comparing segments.
- **Cox Proportional Hazards:** With features. Tells you which features most accelerate or reduce churn risk. Interpretable coefficients.
- **DeepSurv / DeepHit:** Neural network survival models. More accurate than Cox for complex datasets but loses interpretable coefficients.

Sequence Model for Behavioral Data

Instead of aggregated features, use the raw sequence of events: [login, create_doc, invite_teammate] → [login, edit_doc] → [login] → [open_support_ticket] → [?]. Feed this sequence into an LSTM or Transformer. Captures temporal patterns in raw behavior without manual feature engineering.

Step 5: Evaluation

The Right Metrics

- **AUC-ROC:** How well does the model separate churners from non-churners? 0.8 = good, 0.9 = excellent.
- **AUC-PR:** Better than ROC for imbalanced data. If churn rate is 5%, ROC can look good even with a mediocre model.
- **Lift Curve:** What business stakeholders actually care about. If you contact top 20% highest-risk customers, how many churners do you catch vs. random? A 3.25x lift means model is 3.25x better than random.

Time-Based Train/Test Split

Training set: observation windows in Jul-Nov, labels in Aug-Dec. Validation set: observation in Dec, labels in Jan. Test set: observation in Jan, labels in Feb. Always time-based. Never randomly split churn data.

Step 6: Production

The Full Production System

1. Pull last 30 days of behavior for all active customers
2. Compute all features (usage trends, support tickets, etc.)
3. Run churn model — $P(\text{churn})$ score for every customer
4. $\text{Score} > \text{threshold}$: add to intervention queue
5. Prioritize by Expected Value = $P(\text{churn}) \times \text{CLV} \times P(\text{saves}) - \text{Cost}$
6. Top 50 highest-EV customers to Customer Success team, next 200 to email campaign
7. Track outcomes and feed back as new training labels (monthly retrain)

Segmented Thresholds

One threshold does not fit all. High-value customers ($\text{MRR} > 1000$): threshold = 0.15, intervene early. Medium value: threshold = 0.35. Low value: threshold = 0.60, only intervene if very likely to churn. Low-value customers need to be very likely to churn before it is worth spending resources.

Feedback Loop — The Counterfactual Problem

You flag Customer X as high churn risk. Sales team calls them. Customer X stays. Was the model right? Did the call save them? You do not know — this is the counterfactual problem. Solution: Holdout groups. 80% of high-risk customers receive intervention. 20% do not. Compare churn rates. This is an A/B test on your intervention strategy.

Explainability for Customer Success Teams

Customer success managers need to know WHY a customer is flagged. They cannot call and say "our model says you are at risk." SHAP values generate human-readable explanations: (1) Logins dropped 80% vs last month, (2) 4 unresolved support tickets, (3) auto-renew turned off, (4) only 2 features used out of 20. CSM script: "I noticed your team has had some support tickets open — I wanted to check in and make sure you are getting value from the product. I saw usage has been lighter recently..."

How to Answer in an Interview

First clarify how we define churn and the prediction horizon — this affects how we set up training labels and observation windows. Most important technical concern: preventing data leakage. Features must be computed strictly before the label window. For features, the most predictive signals are behavioral trends — not just logins last month but how much has usage changed vs. the previous month. The usage slope feature captures declining engagement elegantly. For the model, start with LightGBM. Add Cox Proportional Hazards to estimate time-to-churn, not just binary risk. Evaluate on AUC-PR and lift curve. In production: daily batch scoring, prioritize interventions by expected value, maintain a holdout group to measure whether interventions actually work, and use SHAP values directly in the customer success dashboard.